

Inside The Race To Build QUANTUM-SECURE NETWORKS

If quantum communication becomes mainstream, our entire approach to digital security could change.



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We handle enormous volumes of information every day. Beyond the data flowing through internet banking, e-commerce, technology, industry, and governance, we constantly exchange personal messages on platforms such as WhatsApp and Facebook. In many ways, the internet has become a vast marketplace of information. Strengthening the security of that information is essential, and this forms the basis of network security.

Satellites anchor much of this communication infrastructure. If an unwanted third party infiltrates a satellite-based system, the accuracy and reliability of the information become uncertain. To understand the risk, it is necessary to examine how the telecommunication system operates and where its vulnerabilities lie.

This telecommunication system is based on a classical computer system, and network security is ensured by the Caesar cipher mechanism, a substitution

encryption technique that shifts each letter in a message by a fixed number of positions. To illustrate, suppose Bimal, from one end of a field, sends a message to Rahim, who is at the other end. Rahim is the only person who knows the code that unlocks the encrypted original message.

For example, the message sent by Bimal says, 'WE MUST GO,' but it is encrypted with a shift of three positions. As a result, the message reaches Rahim as 'ZH PXVW JR.' If Robin attempts to decrypt it midway, he must solve a complex mathematical problem that requires computational assistance. The more complex the code, the more time and high-powered computing capability is needed to break it. This is referred to as cryptographic security.

However, even powerful supercomputers sometimes cannot break this technique, whereas a quantum computer can unravel it almost instantly. This is where quantum cryptography becomes relevant. This security framework is built using

the principles of quantum physics. Quantum cryptography typically works in two ways: 'quantum key distribution' (QKD) and 'quantum entanglement.' Quantum physics protects privacy through multiple mechanisms, one of which is 'quantum measurement.'

Consider a quantum system, such as the state of a photon, a subatomic particle of light, whose state changes when measured. If secret information is stored in a stream of photons, with

